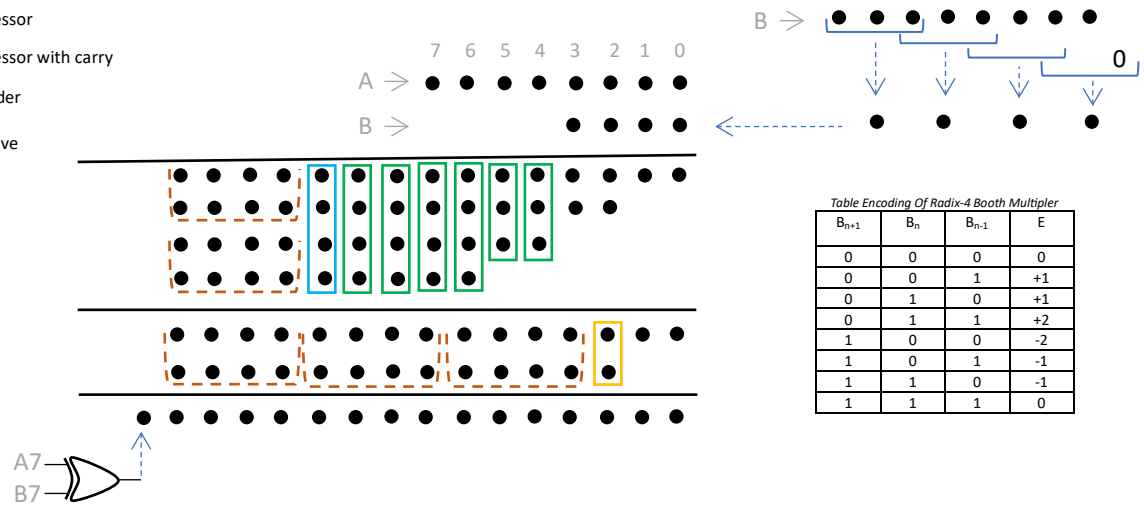


- Approximate compressor
- Approximate compressor with carry
- Approximate half adder
- FPGA CARRY4 primitive



Approximate compressor 3_2:

$$\text{Sum}_0 = P_0 + P_1$$

$$\text{Sum}_1 = P_0 + P_1 \cdot P_2$$

Approximate compressor 4_2:

$$\text{Sum}_0 = P_0 + P_1 + P_2 \cdot P_3$$

$$\text{Sum}_1 = P_0 \cdot P_1 + P_2 + P_3$$

Approximate compressor 4_2 with carry:

$$\text{Sum}_0 = P_0 + P_1 + P_2 \cdot P_3$$

$$\text{Sum}_1 = P_0 \cdot P_1 + P_2 + P_3$$

$$\text{Carry_Out}_0 = P_0 \cdot P_1 \cdot P_2 \text{ [used it in 1}^{\text{st}} \text{ FPGA CARRY4 primitive]}$$

$$\text{Carry_Out}_1 = P_1 \cdot P_2 \cdot P_3 \text{ [used it in 2}^{\text{nd}} \text{ FPGA CARRY4 primitive]}$$

Approximate compressor half adder:

$$\text{Sum}_0 = P_0 + P_1$$

$$\text{Carry}_{\text{Out}} = P_0 \cdot P_1$$